

3.2 Physical chemistry

This section introduces physical chemistry within the general theme of energy.

Learners first learn about the importance of enthalpy changes, their uses and determination from experimental results including enthalpy cycles.

This section then investigates the ways in which a change in conditions can affect the rate of a chemical reaction, in terms of activation energy, the Boltzmann distribution and catalysis.

Reversible reactions are then studied, including the dynamic nature of chemical equilibrium and the influence of conditions upon the position of equilibrium.

Finally, the integrated roles of enthalpy changes, rates, catalysts and equilibria are considered as a way of increasing yield and reducing energy demand, improving the sustainability of industrial processes.

3.2.1 Enthalpy changes

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Enthalpy changes: ΔH of reaction, formation, combustion and neutralisation	
(a) explanation that some chemical reactions are accompanied by enthalpy changes that are exothermic (ΔH , negative) or endothermic (ΔH , positive)	
(b) construction of enthalpy profile diagrams to show the difference in the enthalpy of reactants compared with products	M3.1

- (c) qualitative explanation of the term *activation energy*, including use of enthalpy profile diagrams
- (d) explanation and use of the terms:
- (i) *standard conditions* and *standard states* (physical states under standard conditions)
 - (ii) *enthalpy change of reaction* (enthalpy change associated with a stated equation, $\Delta_r H$)
 - (iii) *enthalpy change of formation* (formation of 1 mol of a compound from its elements, $\Delta_f H$)
 - (iv) *enthalpy change of combustion* (complete combustion of 1 mol of a substance, $\Delta_c H$)
 - (v) *enthalpy change of neutralisation* (formation of 1 mol of water from neutralisation, $\Delta_{\text{neut}} H$)
- (e) determination of enthalpy changes directly from appropriate experimental results, including use of the relationship: $q = mc\Delta T$

Bond enthalpies

- (f) (i) explanation of the term *average bond enthalpy* (as the breaking of 1 mol of bonds in gaseous molecules)
- (ii) explanation of exothermic and endothermic reactions in terms of enthalpy changes associated with the breaking and making of chemical bonds
- (iii) use of average bond enthalpies to calculate enthalpy changes and related quantities (see also 2.2.2 f)

Hess' law and enthalpy cycles

- (g) Hess' law for construction of enthalpy cycles and calculations to determine indirectly:
- (i) an enthalpy change of reaction from enthalpy changes of combustion
 - (ii) an enthalpy change of reaction from enthalpy changes of formation
 - (iii) enthalpy changes from unfamiliar enthalpy cycles
- (h) the techniques and procedures used to determine enthalpy changes directly and indirectly.

M3.1

Activation energy in terms of the minimum energy required for a reaction to take place.

Definitions required for enthalpy changes of formation, combustion and neutralisation only.

Standard conditions can be considered as 100 kPa and a stated temperature, 298 K.

M0.0, M0.2, M2.2, M2.3, M2.4

PAG3

M0.0, M0.2, M2.2, M2.3, M2.4

Formal definition of average bond enthalpy **not** required.

Learners are expected to understand that an actual bond enthalpy may differ from the average value.

M0.0, M0.2, M1.1, M2.2, M2.3, M2.4, M3.1

Definition of Hess' law **not** required. Unfamiliar enthalpy cycles will be provided.

HSW2 Application of the principle of conservation of energy to determine enthalpy changes.

M3.1, M3.2

To include the enthalpy changes covered in 5.2.1 c.

PAG3

HSW4 Opportunities for carrying out experimental and investigative work.